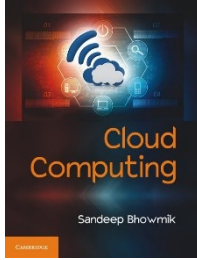


# Cloud Computing

## Chapter 1

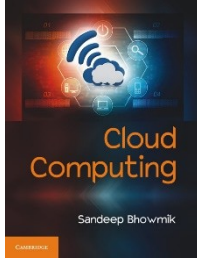
### Introduction

Cambridge University Press



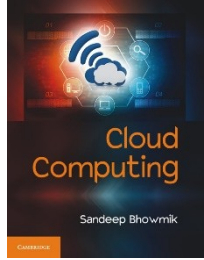
## What is the Buzz About?

- Cloud computing is the next step of accessing web-based services through Internet.
- It provides means for smarter way of doing computing and business.
- Cloud computing became a hot topic only from mid-2008, but the idea is age old.
- The limitations of the traditional computing approaches make the stage ready for its arrival.



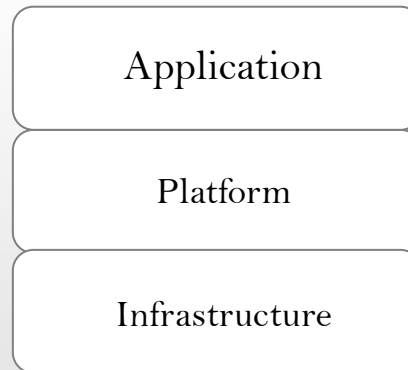
## Limitations of Traditional Computing Approaches

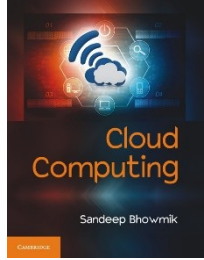
1. Require huge initial investment.
2. System maintenance is the headache of user.
3. Procures hardware gets outdated periodically.
4. Software up-gradation headache.
5. Resource capacity estimation is a tedious task.
6. Resource requirement may vary (increase or decrease) with time.
7. Resource addition/alteration needs complete system shutdown.
8. Less resource utilization results in reduced ROI.
9. For computing, users generally need to have physical access to the system.



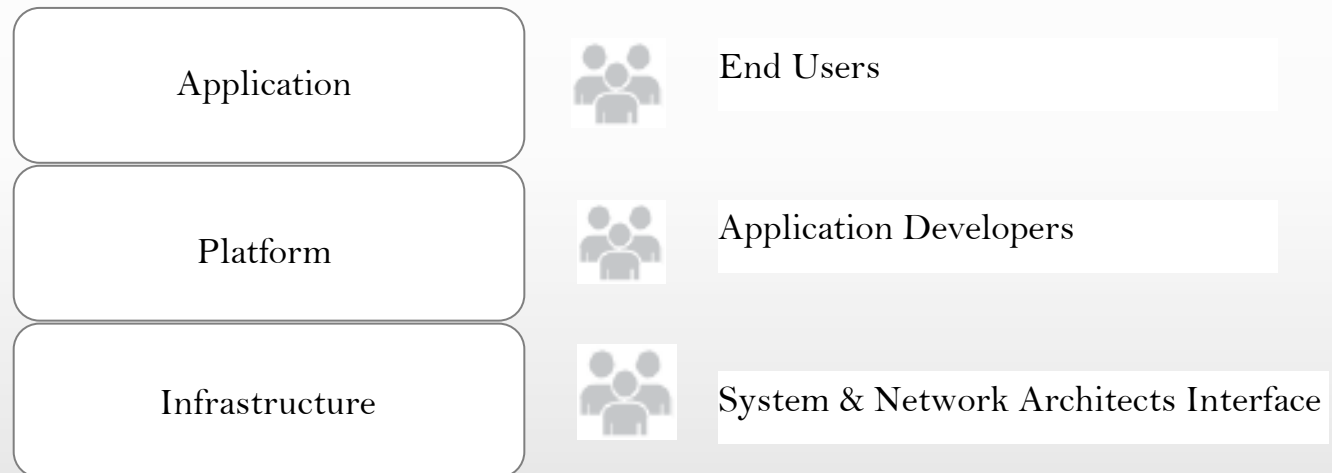
# Three Layers of Computing

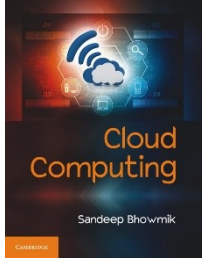
- Infrastructure
- Platform
- Application



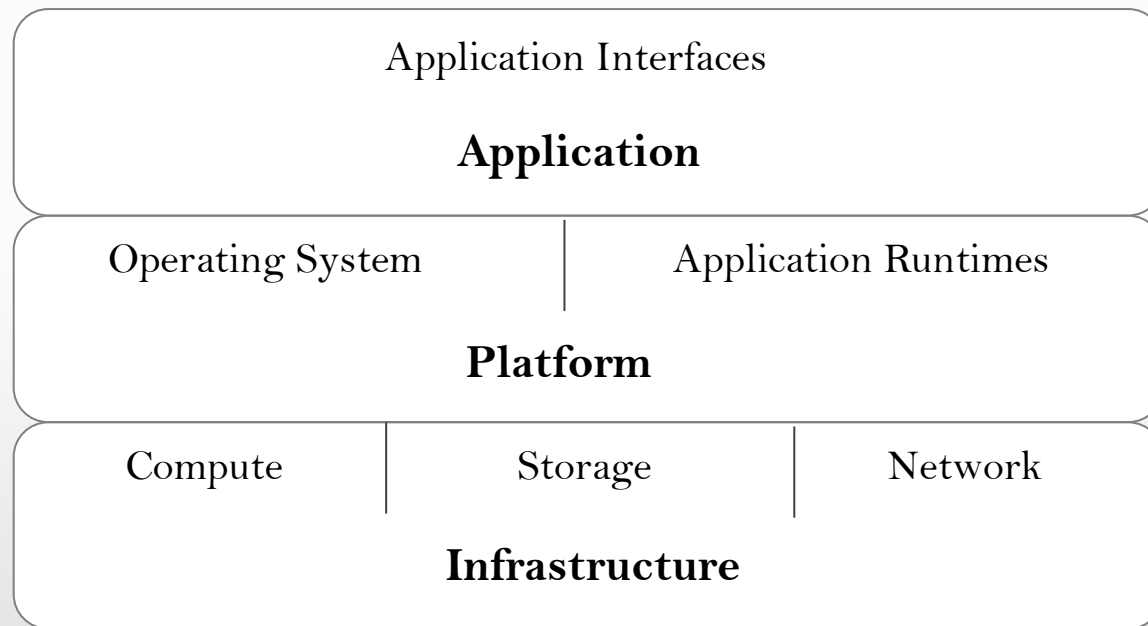


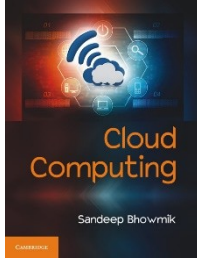
# Users of three different computing layers





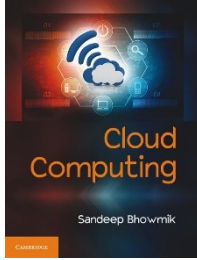
## Different constituents of the three layers





## Three layers in Traditional Computing

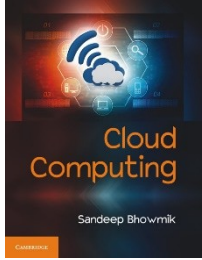
- The boundaries between computing layers were not very clear to general users.
- End users unknowingly had to bear the burden of being concerned about all the three computing layers.
- Subscribers of one computing layer can't fully escape the responsibilities and headaches of arranging or managing the underlying layer(s).



## Computing as Utility Service

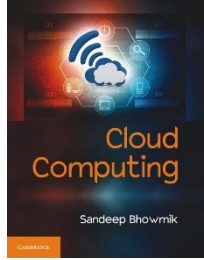
- Utility model provides lots of flexibilities compared to traditional way of computing.
- The three layers of computing, are delivered as utility services in cloud computing model.
  - Infrastructure Service
  - Platform Service
  - Application or Software Service
- Independent Software Vendors (ISV) provide the utility computing services.





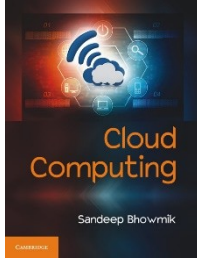
## You are already using Cloud Computing

- E-mail services
- Picture or video sharing services
- Social networking sites



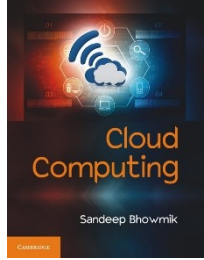
# Influences behind Cloud Service Adoption

- Technological Influences
  - Universal Network Connectivity
  - High-Performance Computing
  - Commoditization



## Influences behind Cloud Service Adoption (Contd.)

- Operational or Business Influences
  - Low-Cost Solution
  - Outsourcing
  - Speed or Responsiveness
  - Automation
  - Small Initial Investment
  - Less Maintenance Cost
  - Mobility
  - Flexibility



**Thank You**